

Review

Translational adaptation in breast cancer metastasis and emerging therapeutic opportunities

Siyu Chen,^{1,2,3,4} Albertas Navickas,^{5,6,*} and Hani Goodarzi^{1,2,3,4,*}

Breast cancer's tendency to metastasize poses a critical barrier to effective treatment, making it a leading cause of mortality among women worldwide. A growing body of evidence is showing that translational adaptation is emerging as a key mechanism enabling cancer cells to thrive in the dynamic tumor micro-environment (TME). Here, we systematically summarize how breast cancer cells utilize translational adaptation to drive metastasis, highlighting the intricate regulation by specific translation machinery and mRNA attributes such as sequences and structures, along with the involvement of tRNAs and other *trans*-acting RNAs. We provide an overview of the latest findings and emerging concepts in this area, discussing their potential implications for therapeutic strategies in breast cancer.

Translational adaptation: a critical and indispensable hallmark of breast cancer metastasis

Breast cancer, the most prevalent cancer worldwide [1], poses a significant public health challenge. A key contributor to cancer-related mortality is metastasis: the process where cancer cells disseminate from their original site to other body parts. During the **metastatic cascade** (see [Glossary](#)) ([Figure 1](#)), cancer cells are challenged by the hostile, highly dynamic, and distinct environments [2]. One of the key strategies for survival and proliferation in these conditions is through translational adaptation. This process, which is distinct from transcriptional regulation, enables cancer cells to rapidly alter protein production in response to environmental stresses (such as low oxygen, limited nutrient availability, immune response), thereby adapting swiftly to the dynamic **TME**.

mRNA translation is a multistep and highly coordinated process and can be broadly divided into three interconnected steps: **translation initiation**, **translation elongation**, and **translation termination** ([Figure 1](#)). Technological progress, notably in ribosome profiling [3] and other RNA sequencing-based approaches, has significantly deepened and expanded our understanding of this intricate process. These advancements unveiled the complex ways in which cancer cells finely adjust their translation machinery during the metastatic cascade. Here, we review the recent findings and emerging concepts in breast cancer metastasis, through the prism of molecular mechanisms of translational adaptation. We focus on how the changes in translation machinery are tailored to specific sequence and structure information encoded in the mRNA and interconnect with additional regulatory layers to drive cancer progression. Our objective is to explore how this growing body of knowledge can lead to novel cancer treatment strategies, particularly those overcoming drug resistance and inhibiting the spread of cancer cells.

Highlights

Translational reprogramming enables swift adaptation of breast cancer cells in the metastatic cascade.

Cancer cells strategically manipulate translation machinery and mRNA attributes to selectively translate mRNAs for specific functional requirements.

tRNAs and other *trans*-acting RNAs significantly shape the translational landscape in the progression of breast cancer.

Advances in comprehending translational adaptation in breast cancer metastasis unveil therapeutic possibilities.

¹Department of Biochemistry and Biophysics, University of California, San Francisco, CA, USA

²Department of Urology, University of California, San Francisco, CA, USA

³Helen Diller Family Comprehensive Cancer Center, University of California, San Francisco, CA, USA

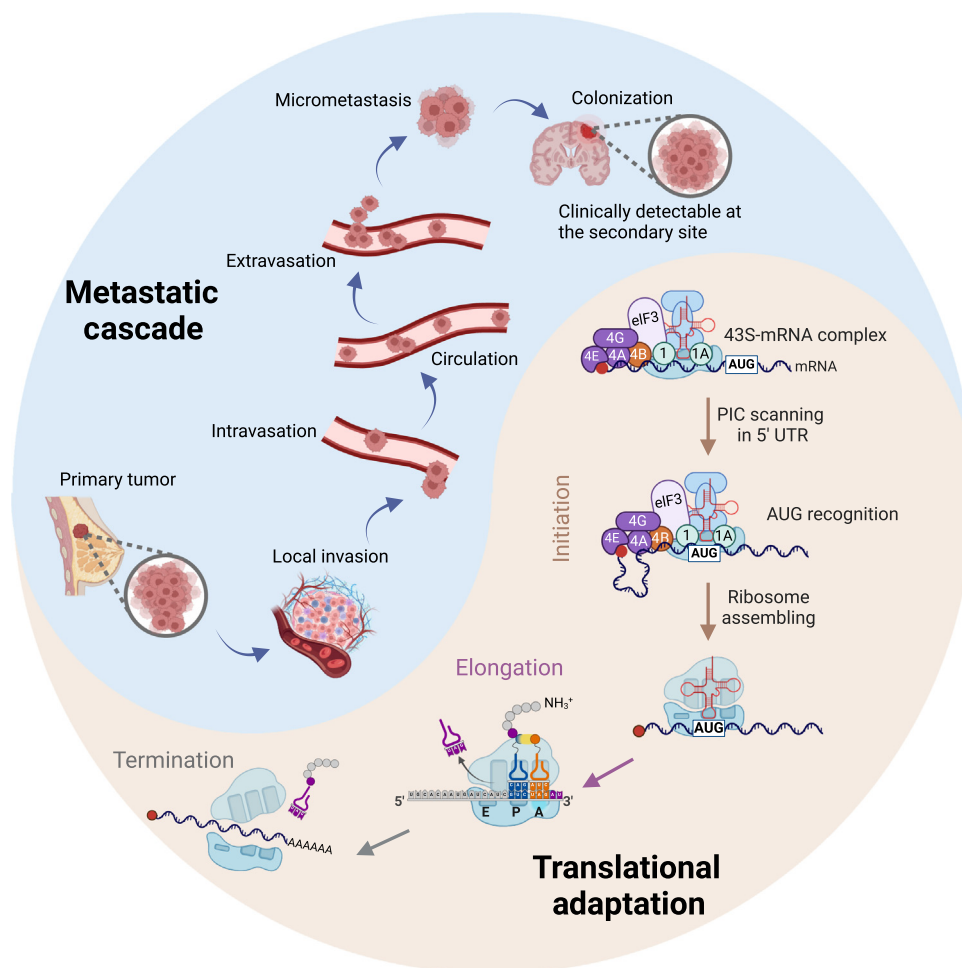
⁴Baker Computational Health Sciences Institute, University of California, San Francisco, CA, USA

⁵Institut Curie, PSL Research University, CNRS UMR3348, INSERM U1278, Orsay, France

⁶Université Paris-Saclay, CNRS UMR3348, INSERM U1278, Orsay, France

*Correspondence: albertas.navickas@curie.fr (A. Navickas) and hani.goodarzi@ucsf.edu (H. Goodarzi).





Trends in Pharmacological Sciences

Figure 1. Translational adaptation during metastatic cascade. The metastatic cascade is a multifaceted process that includes: (i) local invasion, where cancer cells infiltrate nearby tissues post-primary tumor formation; (ii) intravasation, with cancer cells entering the bloodstream or lymphatic system; (iii) circulation, where cells survive and circulate through the body; (iv) extravasation, involving cells exiting the circulatory system at a distant site; (v) micrometastasis, where small cancer cell colonies form; and (vi) colonization, leading to the growth of a detectable metastatic tumor. Central to this cascade is translational adaptation, a multistep process involving initiation, elongation, and termination. At each step of translation, this adaptation is influenced by the translation machinery, alterations in mRNA sequence and structure, and changes in tRNAs and other RNA molecules, crucial for cancer cell survival in dynamic environments. Abbreviations: PIC, preinitiation complex; UTR, untranslated region.

Hijacked translation machinery in the metastatic cascade

The translation process is highly coordinated and orchestrated by hundreds of protein and RNA factors, such as translation factors and ribosomes, along with the precise tuning of information within the mRNA. Given this diversity of molecular components, it is not surprising that cancer cells exploit numerous regulatory and adaptive opportunities within this intricate translation machinery.

Translation factors

Translation initiation is crucial in cancer metastasis, starting with translation initiation factor (eIF) eIF4E binding to the mRNA's 5' cap structure and the formation of the 43S preinitiation complex

Glossary

Aminoacyl-tRNA synthetase

(aaRS): enzymes that attach amino acids to corresponding tRNAs. They ensure accurate amino acid-tRNA pairing, critical for the correct assembly of polypeptides during protein synthesis and for translation fidelity.

Co-translational folding: newly synthesized peptides begin to fold into their specific structures while being threaded through and emerging from the ribosome's peptide exit tunnel.

Metastatic cascade: a multistage process where cancer cells detach from the primary tumor, migrate, and establish metastases in distant organs.

Oncogene: a specific gene with the potential to cause cancer. Mutated or overexpressed proto-oncogenes become oncogenes, leading to abnormal cell proliferation and tumor development. They drive key cancer processes like uncontrolled growth and metastasis.

P-element-induced wimpy testis

(PIWI)-interacting RNAs (piRNAs): typically, 26–32 nt long and a new addition to the ncRNA family, they bind with PIWI proteins and are predominantly found in mammalian germ cells, playing key roles in epigenetic and post-transcriptional regulation.

Ribosome heterogeneity: variability in ribosome composition within cells, resulting from differential ribosomal protein gene expression, rRNA variations, and modifications. This influences selective mRNA translation, impacting the cellular proteome.

RNA G-quadruplex: four-stranded secondary structures formed in G-rich sequences by stacked guanine arrangements.

Synonymous codon: codons encoding the same amino acid. Synonymous codons are used unevenly, which can affect translation efficiency and/or accuracy.

tRNA isoacceptor: tRNA molecules with different anticodons but charged with the same amino acid. They pair with various synonymous codons, influencing translation efficiency and accuracy.

Translation elongation: a protein synthesis phase where ribosomes move along mRNA, adding amino acids to the growing polypeptide chain. This phase is facilitated by the binding of complementary tRNA anticodon sequences to mRNA codons, a key

(PIC). This step is essential for start codon recognition and is facilitated by factors such as eIF4A, eIF4B, and eIF4F, which unwind the mRNA's 5' **untranslated region (UTR)** (Figure 1). Dysregulation of eIF activity, often due to upstream signaling anomalies, has been implicated in breast cancer metastasis, leading to altered translation of mRNAs critical for tumor growth, metastasis, and chemotherapy resistance [4,5]. Particularly, elevated eIF4E levels are significant in predicting cancer progression or relapse [6]. Stress conditions, such as DNA damage or oxidative stress, can cause aberrant phosphorylation of eIF2 α . This change reduces overall translation initiation while enabling adaptive protein synthesis in cancer cells. [7].

Ribosomes

Ribosome, the catalytic hub of the translation machinery, historically seen as a passive, non-selective apparatus comprising rRNAs and ribosomal proteins (RPs), has emerged as a dynamic, heterogeneous, specialized entity in mRNA translation, pivotal in cellular regulation [8]. A significant source of **ribosome heterogeneity** originates from post-transcriptional modifications in rRNA, notably rRNA 2'-O-methylation (Figure 2A). Techniques like RiboMeth-Seq have advanced our understanding of these modifications, revealing their impact on the translation of specific mRNAs and subsequent cellular phenotypes [9]. In the context of breast cancer, rRNA 2'-O-methylation varies both within and between patients, correlating with cancer subtype and tumor grade [10]. This modification, especially in response to MYC **oncogene** expression, influences the selective translation of mRNAs, thus affecting cancer cell behavior and therapy responses [11]. Additionally, the post-translational modifications of RPs contribute to acute translational reprogramming in response to activated signaling pathways. Ribosomal protein S6 (eS6) phosphorylation, modulated by S6K in response to amino acid levels, influences key cancer pathways [8]. Moreover, the emergence of specialized ribosomes, or 'onco-ribosomes', in cancer cells is gaining attention. These ribosomes are tailored to support oncogenic translation programs, modulating cellular functions and metabolic rewiring crucial for breast cancer progression [12].

Tailored mRNA sequence and structure information in facilitating translation reprogramming

The translation process is intricately tuned by the specific sequence and structural information within the UTRs (5' UTR and 3' UTR) and coding sequences of mRNA. This meticulous orchestration of all functional entities of the mRNA body provides a fertile ground for breast cancer cells to adaptively manipulate these attributes of oncogenes and **tumor suppressor genes** to selectively translate mRNAs for distinct functional needs.

5' UTR structure

In breast cancer cells, RNA structural and sequence changes within the 5' UTR significantly impact translation initiation. These alterations influence ribosome entry, scanning, and start codon recognition, crucial for cancer cell proliferation and metastasis [5]. The iron-responsive element (IRE), a classic and highly conserved 5' UTR structural element, regulates iron homeostasis by controlling translation initiation. In breast cancer, increased iron regulatory protein 2 (IRP2) expression enhances IRP2's binding to the IRE in the ferritin H (FTH) 5' UTR, thereby reducing FTH translation [13]. This underscores the pivotal role of IRE-mediated translation control in cancer progression and chemosensitivity [14]. Additionally, **RNA G-quadruplexes** in the 5' UTR affect mRNA translation (Figure 2B), and compounds like RGB-1 targeting these structures to reduce oncogene expression, offering a therapeutic strategy [15,16].

The internal ribosome entry site (IRES), characterized by its unique secondary or tertiary RNA structure, is vital for cap-independent translation of specific mRNAs in cancer progression and

process for ensuring the accuracy and efficiency of protein synthesis, persisting until a stop codon is encountered.

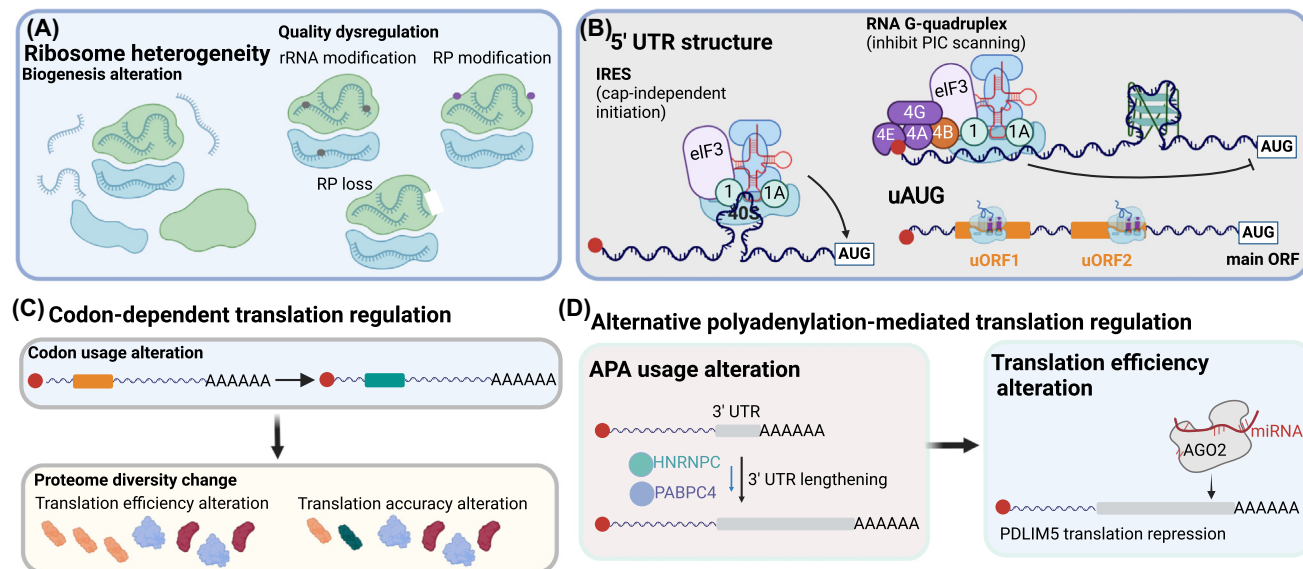
Translation initiation: the first phase of protein synthesis where the ribosome assembles at the mRNA's start codon. This step is widely recognized as the rate-limiting step in translation and is critical for accurate and efficient protein translation.

Translation termination: the final phase of protein synthesis, initiated when a stop codon enters the ribosome's A site. It involves release factors that recognize the stop codon and results in the release of the newly synthesized polypeptide from the ribosome and ribosome dissociation.

Tumor microenvironment (TME): a dynamic ecosystem surrounding cancer cells, consisting of immune and stromal cells, extracellular matrix, vasculature, and signaling molecules. It evolves under stress and influences cancer progression, particularly in transforming normal cells into malignant forms and promoting their proliferation, invasion, and metastasis.

Tumor suppressor gene: acts as crucial brakes on cell proliferation and cancer development.

Untranslated region (UTR): non-protein-coding sequences at both ends of mRNA. The 5' UTR regulates translation initiation, ribosome assembly, and start codon recognition. The 3' UTR is involved in translation termination, mRNA stability, and localization. Both are crucial for efficient mRNA functionality.



Trends in Pharmacological Sciences

Figure 2. Influences of translation machinery and mRNA attributes on translational adaptation processes. (A) Sources of ribosome heterogeneity. Ribosome heterogeneity arises from a variety of sources, including variations in ribosome biogenesis, the incorporation of rRNAs and ribosomal proteins (RPs), and aspects of quality control. This latter category encompasses modifications of rRNA and RP, as well as RP loss, each contributing to the diversity in ribosome structure and function. (B) Regulatory structures and elements in 5' untranslated region (UTR) and their impacts on translation initiation. The RNA structures within the 5' UTRs, such as RNA G-quadruplexes, can act as physical barriers to the unwinding activities of the scanning ribosome and associated eukaryotic initiation factors. Internal ribosome entry sites (IRESs) facilitate an alternative mechanism of translation initiation, enabling ribosome recruitment independently of the mRNA cap structure. Upstream open reading frames (uORFs) and upstream AUGs (uAUGs) generally act to repress the translation of the main downstream ORF. (C) Codon usage and its effects on translation elongation. The efficiency and accuracy of translation can be modulated by codon usage, which in turn contributes to proteome diversity. (D) Influence of alternative polyadenylation (APA). Changes in polyadenylation sites, specifically the proximal-to-distal APA switch, can significantly impact translation efficiency and are linked to variations in metastatic potential. Abbreviation: PIC, preinitiation complex.

metastasis (Figure 2B). Cellular IRESs, activated under stress, influence the production of factors like HIF1A, thereby enhancing tumor angiogenesis and hypoxia responses in breast cancer [5]. Identification of IRES-dependent events in metastasis and therapeutic resistance requires stringent validation to mitigate false positives, despite growing evidence supporting their implication [17].

Regulatory elements in 5' UTR

Approximately 40% of human mRNAs harbor upstream open reading frames (uORFs) in their 5' UTRs, pivotal in stress-responsive gene expression and cell fate determination under stress [18]. uORF-containing genes, enriched in oncogenes, impede the downstream translational regulation of the primary protein coding sequence [19] (Figure 2B). Not limited to translation suppressive roles, some uORFs encode biologically active peptides, exemplified by uPEP2 in kinase PKC- η , which reduces breast cancer cell survival and migration while boosting chemotherapy efficacy [18].

The accurate and efficient recognition and selection of the translation initiation site on mRNAs are essential for proper protein production [20]. Inefficient start codon recognition can result in 'leaky scanning', where part of the PIC continues scanning and initiates translation at downstream sites. The proto-oncogene Scribble (SCRIB), which functions as a tumor suppressor in key signaling pathways like mitogen-activated protein kinase and extracellular signal-regulated kinase (MAPK/ERK) and phosphoinositide 3-kinase (PI3K)/AKT/mechanistic target of rapamycin (mTOR) exemplifies this. The translation efficiency of SCRIB is contingent on the leaky scanning frequency of a translatable overlapping ORF, oSCRIB, serving as a regulatory mechanism in

normal cells to limit SCRIB expression, and is associated with the survival and proliferation of cancer cells [21].

Sequence information in coding regions

Synonymous mutations in coding regions are prevalent in cancers and can drive cancer progression without altering the protein sequence [22,23]. Substituting codons with their synonymous counterparts can perturb the balance between **synonymous codon** usage and the abundance of the respective **tRNA isoacceptors** [24]. This can lead to altered mRNA stability and translation efficiency and accuracy [25], potentially resulting in changes in protein abundance, folding, and function [26–28]. Studies have shown that codon usage not only regulates KRAS expression and oncogenesis [29] but also impacts the conformation of the KRAS protein, potentially through **co-translational folding** mechanisms [30]. These insights into codon-dependent translational control can be developed to target and rectify translation anomalies in breast cancer cells (Figure 2C). This approach could potentially offer more precise and effective treatments for breast cancer, leveraging the nuanced understanding of codon influence on protein synthesis and function.

Regulatory elements in 3' UTR

The 3' UTR often harbors regulatory elements that interact with various *trans*-acting factors, such as miRNAs, their associated machinery, and RNA-binding proteins (RBPs). These interactions can significantly modulate translation in breast cancer cells. Aurora Kinase A (AURKA), an oncogenic kinase with pivotal roles in mitosis, also contributes to cancer-promoting activities in cell proliferation, survival, migration, and cancer stem cell phenotypes [31]. A recent study found that triple-negative breast cancer samples showed an increased short/long ratio of AURKA 3' UTR isoforms. The shorter isoform, lacking the hsa-let-7a miRNA binding site found in the longer variant, demonstrated higher translational efficiency. This miRNA targeting is crucial for the cell cycle-dependent translational regulation of the AURKA long 3' UTR isoform, influencing the oncogenic activation of AURKA in cancer cells [32]. Our recent study observed that the 3' UTR lengthening of a subset of metastasis-suppressing genes affects metastatic ability [33] (Figure 2D). In highly metastatic breast cancer cells, heterogeneous nuclear ribonucleoprotein C (HNRNPC) deficiency affects proximal-to-distal poly(A) site switch and causes translational downregulation of a subset of metastasis-inhibiting genes, such as PDLIM5, leading to enhanced metastatic potential [33].

RNA modification

RNA modifications are considered as chemo-structural features of RNA molecules. N^6 -methyladenosine (m^6A) is the most common modification in mRNAs, playing a pivotal role in regulating mRNA fate by its readers [34]. Recent studies have highlighted its significant impact on translation control in cancer biology. m^6A modification has been identified as a facilitator of cap-independent translation in cancer, altering the translational efficiency of specific mRNAs and driving cancer progression [35]. m^6A deposition is catalyzed by a methyltransferase complex, with methyltransferase-like 3 (METTL3) as its key catalytic subunit. Recent evidence indicates that METTL3 plays vital roles in both oncogenic and tumor-suppressive functions [36]. The regulation of METTL3 through acetylation impacts its localization and, consequently, breast cancer metastasis. A METTL3 mutant mimicking acetylation status alters translation and reduces metastatic potential, suggesting new therapeutic avenues in breast cancer treatment [37]. Other RNA modifications, such as m^5C , m^1A , and pseudouridine, have also exhibited regulatory effects on translation in cancer [38]. This growing understanding of the role of RNA modification in translational adaptation underscores its potential as a therapeutic target in breast cancer treatment.

tRNAs and other *trans*-acting RNAs: shaping altered translational landscapes in metastasis

A considerable part of the human genome is involved in producing RNAs but is not classified as protein-coding mRNAs [39]. These noncoding RNAs (ncRNAs), which include tRNAs, tRNA-derived fragments (tRFs), long noncoding RNAs (lncRNAs), circular RNAs (circRNAs), and **P-element-induced wimpy testis (PIWI)-interacting RNAs (piRNAs)**, have been increasingly recognized for their pivotal roles in regulating breast cancer metastasis [39]. Recent advancements highlight the emerging significance of these ncRNAs in the translational regulation, introducing new layers of complexity to our understanding of breast cancer progression.

tRNA and aaRS

In recent years, the tRNA molecule has emerged as a dynamic regulator of gene expression [40]. Increasing evidence indicates that the tight regulation of translation elongation in breast cancer cells may also be attributed to alterations in tRNA availability (Figure 3), which can be caused by genetic changes, transcriptional dysregulation, and alterations in tRNA stability [40]. This includes both the abundance of tRNAs and their specific amino acid charge, factors that are associated with translation efficiency and accuracy during the decoding process [25,41]. Variations of tRNA availability can lead to deregulated protein production and impact oncogenic signaling pathways. We previously found that aggressive breast cancer cells exhibited upregulated expression of specific tRNAs, tRNA^{Glu}_{UUC} and tRNA^{Arg}_{CCG}. These upregulated tRNAs promote the translation of certain mRNAs in a codon-dependent manner, particularly affecting genes like EXOSC2 and GRIPAP1, further enhancing the invasiveness and metastatic ability of breast cancer cells [42]. This study indicates that specific tRNAs work as regulators of translation efficiency

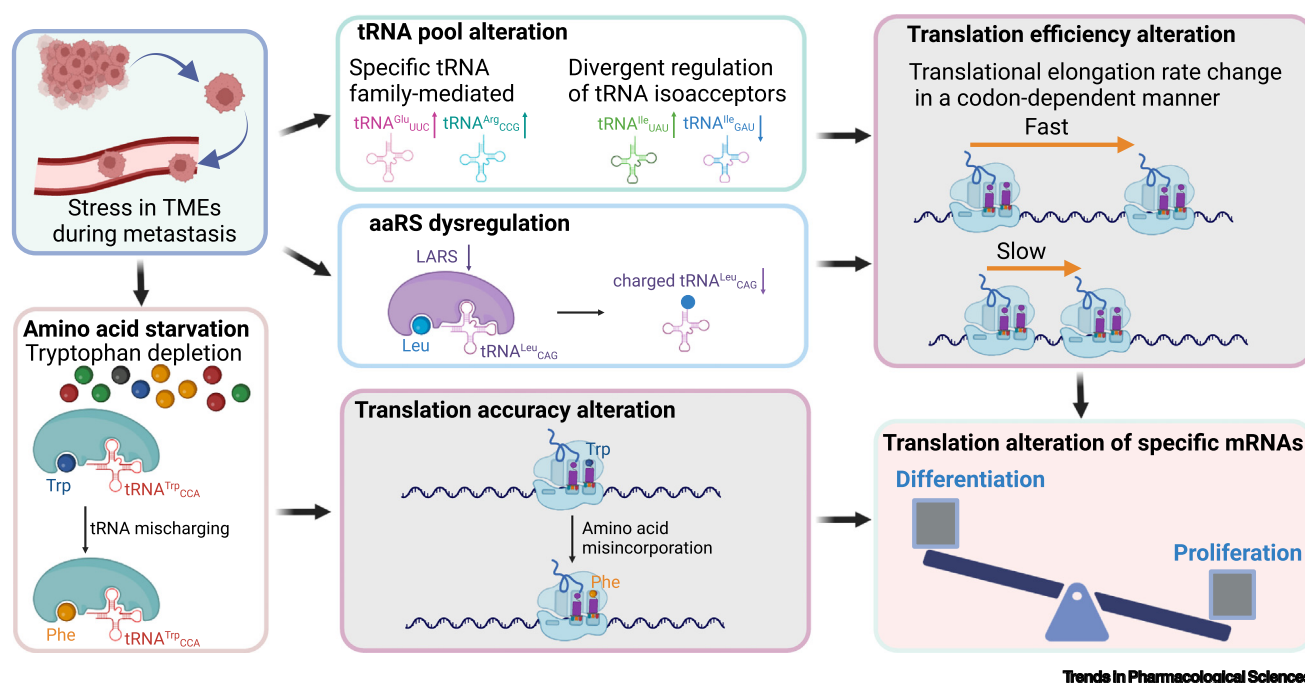


Figure 3. The role of tRNA and aminoacyl-tRNA synthetase (aaRS) in translational adaptation. Under stress conditions, alterations in the tRNA pool and dysregulation of aaRS can occur, along with the mischarging of tRNAs (charging tRNAs with incorrect amino acids). These changes can significantly affect translation efficiency and accuracy, leading to selective dysregulation of specific mRNAs. This dysregulation caters to the particular functional requirements of cancer cells. Abbreviation: TME, tumor microenvironment.

to drive breast cancer progression. Another interesting example is the opposing effects of two tRNA isoacceptors decoding synonymous codons: tRNA^{Lys}_{AAA} enhances, whereas tRNA^{Lys}_{AAG} inhibits metastatic colonization in animal models [43]. These tRNAs divergently affect the translation of genes promoting cell growth, aligning with the genomic abundance or scarcity of their respective codons in cell division-related genes [44]. Breast cancer cells utilize specific tRNA signatures driving the transition towards a more aggressive, growth-favoring state (Figure 3). These results indicate that tRNA availability alteration can shape translational landscapes in cancer cells and represents a critical aspect of protein synthesis control in breast cancer metastasis.

tRNA molecules are critical to accurate protein synthesis, interfacing with **aminoacyl-tRNA synthetases (aaRSs)** that play a crucial role in this process. aaRSs are responsible for attaching specific amino acids to their corresponding tRNAs, a process known as aminoacylation. The fidelity of mRNA translation is heavily reliant on this process. Stringent quality control mechanisms support the aminoacylation, ensuring that codon reassignment errors are avoided and the integrity of protein synthesis is maintained.

Cancer cells utilize their translation machinery to cope with challenges like nutrient and oxygen scarcity, often arising from rapid proliferation. A notable example is the adaptation to tryptophan depletion in cancer cells, where they employ tryptophan-to-phenylalanine codon reassignment (W>F), triggered by an aminoacylation error by tryptophanyl-tRNA synthetase (WARS1). This W>F switch, associated with compromised protein function, plays a role in oncogenic signaling and tumor immunoreactivity [45]. Furthermore, leucyl-tRNA synthetase 1 (LARS1) functions as an intracellular leucine sensor, initiating the mTOR signaling pathway and contributing to translational dysregulation in cancer [46]. In an intriguing study, Passarelli *et al.* found that breast cancer cells can selectively downregulate LARS1 to modulate specific mRNAs. This reduction in LARS1 decreases the availability of particular leucine tRNA isoacceptors, leading to a codon-dependent inhibition of protein synthesis and affecting growth-suppressive genes such as EMP3 and GGT5 [47]. These findings underscore the role of aaRSs in cancer-specific modulation, selectively impacting translation elongation during breast cancer progression (Figure 3).

tRF

tRFs, small RNA molecules originating from mature or precursor tRNAs, are emerging as significant regulators in breast cancer progression. In breast cancer, hypoxia-induced fragmentation of specific tRNAs, notably tRNA(Glu), tRNA(Asp), tRNA(Gly), and tRNA(Tyr), plays a crucial role in destabilizing multiple oncogenic transcripts [48]. This suppression is achieved by displacing the RBP YBX1 from target mRNA 3' UTRs through sequence-specific interactions, leading to the destabilization of oncogenic transcripts and impacting their translation. Subsequently, this results in reduced growth, invasion, and metastasis of breast cancer cells [48]. In this case, tRFs act similarly to miRNA, the well-described regulators of mRNA stability and translation, which have been extensively reviewed elsewhere, including their role in metastasis [49].

In oxidative stress conditions, breast cancer cells show targeted fragmentation of tRNA^{Tyr}_{GUA}, leading to its depletion and affecting translational control in a codon-dependent manner. This depletion hinders the effective translation of tyrosine-rich genes crucial for cell growth and metabolism [50]. These findings highlight the intricate role of tRFs in modulating oncogenic processes, offering insights into the complex interplay between tRNA stability, translation control, and breast cancer progression.

lncRNA

lncRNAs, a class of ncRNAs longer than 200 nucleotides and forming a main source of ncRNAs, serve diverse roles in cancer cell biology. A notable example is metastasis-associated lung

adenocarcinoma transcript 1 (MALAT1), which is known for its involvement in metastasis and drug resistance [51]. Recent evidence, however, increasingly highlights the regulatory role of lncRNAs in translation, expanding our understanding of their function in breast cancer. One such lncRNA, PYCARD-AS1, specifically binds to the 5' end of PYCARD mRNA in breast cancer cells. This interaction impedes ribosome assembly on PYCARD mRNA, effectively inhibiting its translation [52]. Additionally, recent observations reveal that lncRNAs, such as NR2F1-AS1, can regulate cancer metastasis via IRES-mediated translation, influencing dormancy in breast cancer [53]. Beyond their established regulatory roles, recent research has uncovered that lncRNAs can also encode peptides crucial in breast cancer progression. The newly identified peptide GT3-INCP, derived from lncRNA LINC00992, interacts with GATA3 protein to coregulate gene expression linked to estrogen response and cell proliferation. The altered expression of GT3-INCP is associated with poor prognosis [54]. These discoveries highlight the complex and significant role of lncRNAs in translation control and underscore their potential as therapeutic targets in breast cancer.

circRNA

circRNAs, produced from back-splicing of exons of precursor RNAs, were thought of as ncRNAs because they are devoid of a 5' end cap and a 3' end poly (A) tail necessary for cap-dependent translation [55]. But recent studies have demonstrated that a subset of circRNAs can be translated using mechanisms like IRES or m⁶A-dependent initiation, influencing the expression of proteins associated with survival and angiogenesis in cancer [56]. The discovery of circRNAs such as circFGFR1p, originating from the Fibroblast Growth Factor Receptor 1 (FGFR1) locus, highlights their ability to regulate critical cancer-related pathways [57]. Specifically, circRNA-encoded circFGFR1p serves as a negative regulator by inhibiting the expression of the FGFR1 oncoprotein. However, in cancer cells, there is a noticeable reduction in circFGFR1p expression, suggesting that the loss of this regulatory mechanism may contribute to the FGFR1 overexpression and cancer progression. Intriguingly, circRNAs have emerged as promising platforms for gene expression due to their stability and protein-encoding potential from both synthetic and some endogenous forms [58,59]. These findings and advancements open new possibilities for circRNA-based targeted breast cancer therapies.

In addition, piRNAs are emerging as non-negligible regulators in mRNA dynamics, influencing mRNA stability and translation [60,61], with their dysregulated expression implicated in cancer progression [62]. In breast cancer cells, research suggests that the interaction of piR-36712 with SEPW1P RNA may facilitate the targeting of SEPW1 RNA by miR-7 and miR-324, potentially enhancing mRNA degradation and reducing translation, further impacting cancer progression [63,64]. This points to a potential role for piRNA in regulating translation during the metastatic cascade, underscoring the need for further investigation into the therapeutic possibilities of piRNA in breast cancer treatment.

Tuned spatiotemporal regulation is emerging as a new dimension in translational adaptation

Breast cancer progression is significantly influenced by the TME. Understanding cancer cell heterogeneity offers insights into the dynamics of TME, particularly through spatial and temporal regulation studies.

Spatial translation regulation

Currently, research on spatial translation regulation mainly focuses on subcellular localization. Cancer cells, in response to acute extracellular stress during epithelial-to-mesenchymal transition (EMT) and survival under chemotherapeutic and radiotherapeutic treatments, often relocate

specific mRNAs to stress granules to transiently halt translation [65]. Inhibiting Ras-GAP SH3 domain binding proteins (G3BP1/2), crucial for stress granule formation and function, has been shown to impede breast cancer cell migration and reduce distant metastasis [66,67].

Throughout cancer progression, RNA molecules are strategically localized to specific subcellular regions via interactions with RNA binding proteins and RNA regulatory elements (Figure 4A). One study noted that during EMT in breast cancer, LARP6 expression increases, leading to the relocalization of RP-mRNA to actin-rich cell protrusions. This relocalization facilitates the preferential translation of selected mRNA subsets, supporting invasive growth [68].

The recent development of ribosome-bound mRNA mapping (RIBOmap) by Zeng *et al.* presents a novel 3D imaging technique to profile actively translating mRNAs at single-cell and subcellular levels with spatial coordinates [69]. This technique uses a three-probe system to target specific mRNAs, allowing for the precise mapping of mRNA translation *in situ* (Figure 4B). Such advanced methodologies are crucial in enhancing our understanding of spatial translation regulation at both single-cell and spatial resolution, offering new insights into the complex mechanisms of breast cancer progression and metastasis.

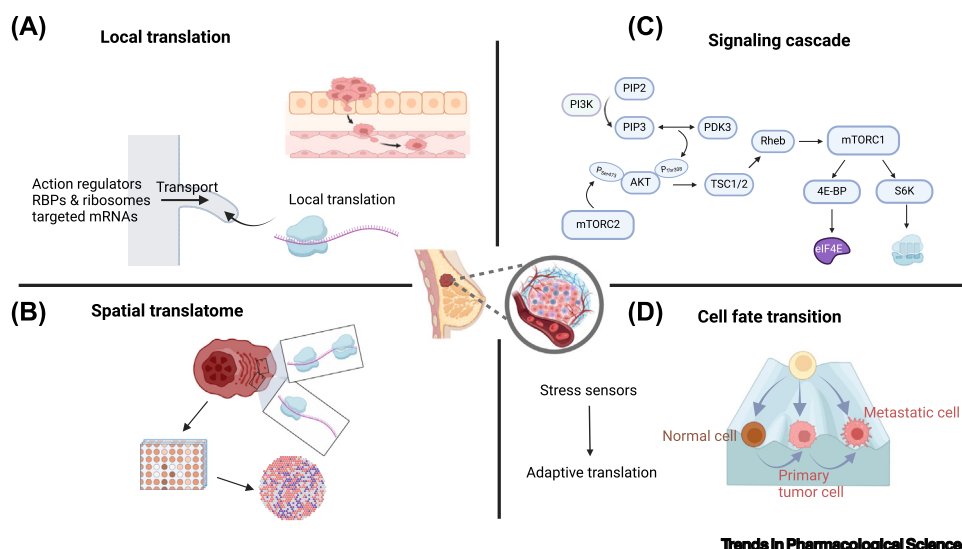


Figure 4. Spatial and temporal translation regulation in dynamic tumor microenvironments (TMEs). Spatial translation regulation in (A) subcellular translation and (B) spatial translatoeme. (A) Subcellular translation. During metastasis, mRNA molecules and translation machinery undergo strategic relocalization to specific subcellular regions. This targeted relocalization facilitates the selective translation of specific mRNA subsets, crucial for supporting invasive growth and enhancing metastatic potential. (B) Spatial translatoeme. This technique provides deep insights into the spatial dynamics of TME in metastasis by systematically measuring and mapping translation activity, thereby elucidating the locations of specific translation processes. Temporal translation regulation in (C) signaling cascade and (D) cell state transition. (C) Signaling cascade. In response to changes in the TME, cancer cells adapt by selectively translating mRNAs, coordinated through a temporal interplay of various oncogenic signals. This selective translation is key in modulating the metastatic potential of the cells. Activated phosphoinositide 3-kinase (PI3K) contributes to mechanistic target of rapamycin (mTOR) activation, which in turn induces the phosphorylation of ribosomal protein S6 kinase (S6K) and 4E-BP, an inhibitor of the translation initiation factor eIF4E. The result is an increased production of ribosomes and enhanced protein synthesis, crucial factors in breast cancer progression. (D) Cell state transition. Gaining a comprehensive understanding of translational reprogramming during cancer cell state transitions is essential. It involves exploring how translation regulation responds and adapts to different stresses and phases during metastasis. Abbreviations: PDK3, pyruvate dehydrogenase kinase 3; PIP2, phosphatidylinositol 4,5-bisphosphate; PIP3, phosphatidylinositol (3,4,5)-trisphosphate; RBP, RNA-binding protein; TSC, tuberous sclerosis complex.

Temporal translation regulation

Breast cancer metastasis progresses through a well-defined temporal sequence, initiated by aberrant signaling and disordered gene expression. This process involves a stepwise adaptation of cancer cells to the TME, leveraging the translational machinery to respond to oncogenic signals at specific timing. The signaling cascade is a well-explored example. Mutations in the PIK3CA gene, a frequent occurrence in primary breast cancer, lead to PI3K activation [70]. Then, activated PI3K contributes to mTOR activation and, finally, abnormally activated mTOR in cancer cells can increase the production of ribosomes and enhance protein synthesis that promotes tumor growth, metastasis, and invasion (Figure 4C) [71].

Additionally, the dynamic between cancer cells and the host's immune system also demonstrates temporal regulation, progressing through immune elimination, equilibrium, and escape stages. During these phases, cancer cells fine-tune the translation of surface ligands to circumvent immune detection, enabling unrestrained proliferation and invasion (Box 1).

However, our current understanding lacks a detailed temporal map of translational reprogramming across the metastatic cascade, especially regarding cancer cell state transitions (Figure 4D). Such knowledge could unveil new insights into the timing and pathways of metastatic spread, as well as the development of therapeutic resistance. Furthermore, integrating this understanding with spatial contexts promises a more holistic view of breast cancer progression. It would account for the complex dynamics within the TME and how these intricacies influence cancer behavior. This approach could lead to more targeted and effective strategies for managing and treating breast cancer.

Box 1. Translational reprogramming in immune evasion

Immune evasion, a known hallmark of cancer metastasis [86], is increasingly recognized as being influenced by translation regulation [87]. Breast cancer cells exploit this mechanism to escape immune surveillance, offering a promising avenue for cancer immunotherapy development.

Breast cancer cells bypass immune recognition by modifying their surface ligands, which interact with receptors on immune effector cells. This intrinsic mechanism of immune evasion is exemplified by the alteration of immune checkpoint pathways, such as programmed death-ligand 1 and programmed death-1 (PD-L1/PD1) and major histocompatibility complex class II and lymphocyte-activation gene 3 (MHC-II/LAG3), during the metastatic cascade [88]. Notably, PD-1 and PD-L1 play a pivotal role in maintaining immune tolerance in the TME [89]. The expression of PD-L1, primarily regulated by the IFN γ -activated Janus kinase/signal transducer and activator of transcription (JAK/STAT) pathway [90,91], is subject to translational control. Recent studies have shown that the antagonism of translational repression of uORFs in JAK2 mRNAs by density regulated re-initiation and release factor (DENR) can decrease PD-L1 expression, thereby impeding tumor growth [92]. Additionally, Xu *et al.* reported the overtranslation of PD-L1 in the metastatic cascade, caused by bypassing uORF-mediated translation repression [93]. Beyond these, a recent study observed that circFAM53B, a tumor-specific circRNA, was discovered to induce cryptic antigenic peptides in human breast cancer [94]. The expression of these peptides is correlated with patient outcomes, highlighting their potential to enhance antitumor immunity. Vaccines derived from these tumor-specific circRNAs or peptides have shown promising results in mouse models, particularly in boosting the infiltration of tumor-antigen-specific cytotoxic T cells [59,95]. These findings underscore the potential for exploiting translational regulation in devising breast cancer immunotherapies.

Furthermore, translation regulation is crucial in the maturation and homeostasis of immune cells. Studies indicate that the translation of mRNAs encoding RPs is upregulated during T cell expansion and downregulated in the contraction phase [96]. Furthermore, naive T cells, characterized by idle ribosomes and repressed mRNAs, rapidly activate these components upon stimulation, underscoring the dynamic nature of translation in the immune response [97]. The critical role of eIF5A activity in the transition of primary T cells from a naive to an activated state also emphasizes the importance of translation in immune functions [98]. These insights provide a foundation for new strategies in immunotherapy, targeting the intricate translation mechanisms within breast cancer and immune cells.

Advancements in understanding translational adaptation during breast cancer metastasis unlock therapeutic potentials

The interplay between translational adaptation and breast cancer metastasis uncovers promising therapeutic opportunities. In this section, we delve into the multifaceted therapeutic potentials resulting from the advances in translational adaptation during breast cancer metastasis. Our aim is to pave the way for applying molecular insights to achieve tangible clinical outcomes.

Therapeutic opportunities stemming from hijacked translation machinery

Translation factors, especially eIF4A and eIF4E, have been extensively studied in breast cancer treatment, with recent reviews highlighting their potential in combating cancer metastasis [5,72]. Intriguingly, beyond targeting translation factors, the dynamic roles of ribosomes in cancer cells open up new avenues for therapeutics, particularly targeting ribosome biogenesis. Metarrestin, targeting the metastasis-linked perinucleolar compartment, disrupts nucleolar structure and inhibits RNA polymerase I, effectively suppressing metastasis and presenting a new therapeutic approach in breast cancer treatment [73]. Additionally, the modification of rRNA, such as 2'-O-methylation, presents novel opportunities in future breast cancer prognostics and treatment [10].

Therapeutic opportunities arising from tailored mRNA sequence and structure analyses

Cancer cells' ability to tailor mRNA sequence features for modulating translation plays a critical role, presenting selective vulnerabilities for therapeutic targeting in breast cancer. IRES-mediated translation, a key factor in malignant progression and treatment resistance, emerges as a promising target for breast cancer therapy. Metformin, an antidiabetic medication, also acts as an anticancer agent by enhancing IRES-dependent translation of antiapoptotic proteins through mTORC1 inhibition in breast cancer cells [74]. Recently, Cao *et al.* developed translation-activating RNAs (taRNAs), bifunctional RNA molecules combining a guide sequence targeting specific 3' UTR sites of an mRNA with an effector domain derived from IRESs to activate translation of targeted mRNAs *in trans* [75]. This approach results in an RNA molecule capable of specifically enhancing the translation of a broad spectrum of targeted mRNAs in mammalian cells. Beyond targeting 5' UTR attributes, our recent study has identified the potential of targeting 3' UTR length in breast cancer therapy, where the small molecule T4, by shortening 3' UTR of a subset of mRNAs, showed efficacy in repressing breast cancer metastasis [33]. Thus, these specific mRNA attributes offer promising targets for breast cancer treatment.

Furthermore, the application of engineered mRNA molecules in cancer treatment is gaining widespread attention. A prime example is mRNA medicine, which has garnered significant interest following the successful deployment of mRNA vaccines against coronavirus disease 2019 (COVID-19). The ability to rapidly identify mRNA therapeutic candidates, coupled with the relatively modest infrastructure requirements for mRNA medicine production, enables the swift development of treatments ranging from vaccines for the general population to personalized cancer therapies [76]. Central to the efficacy of these mRNA medicines is the optimization of the translation process, which hinges on a deep understanding of the mRNA sequence features that govern translational efficiency and accuracy. Key design aspects like codon usage, efficient and accurate translation initiation, and mRNA structure are critical for maximizing therapeutic outcomes [76,77]. Insights into the nuances of mRNA translation can thus drive the development of personalized mRNA medicines, tailored to meet the unique molecular demands of individual cancer treatments.

Therapeutic opportunities unveiled by tRNAs and other *trans*-acting RNAs

The therapeutic potential of tRNAs is increasingly being explored in the treatment of various diseases, suggesting promising applications in breast cancer therapy. Delivering a wild-type pre-tRNA^{Phe} into cells from patients with a mitochondrial tRNA^{Phe} mutation, using lipid nanoparticles,

has been shown to restore mitochondrial function [78]. This finding opens the door to the potential use of tRNA-based treatments in breast cancer. Moreover, engineered tRNAs present a novel approach to address translational challenges. By introducing the correct amino acid at mutation sites, engineered tRNAs can enable read-through of premature termination codons, adjust the reading frame, and restore functional peptide sequence [79,80]. These developments highlight the versatility of tRNA molecules as therapeutic tools in potentially rectifying aberrant protein synthesis in breast cancer cells. Additionally, aaRSs are emerging as novel therapeutic targets [81,82], exemplified by GARS, which suppresses tumor growth *in vivo* by activating the ERK pathway in cancer cells [82], and the increasing attention to LARS due to its role in leucine sensing, regulation of the mTOR pathway, and growth-suppressive gene expression [46,47]. Efforts to develop LARS inhibitors are showing promise as anticancer drugs [83]. Furthermore, circRNAs are gaining attention as viable alternatives to mRNA-based vaccines, with their stability indicating broader applications for RNA-mediated therapies [59]. In addition, peptides derived from lncRNAs have been shown to elicit a strong immune response, leading to a notable delay in tumor growth. This suggests the potential of lncRNA genes in encoding immunogenic peptides, which could be harnessed for cancer vaccine development [84].

Therapeutic potentials derived from the insights of spatiotemporal regulation

The evolving understanding of spatiotemporal translation adaptation offers new therapeutic opportunities in breast cancer diagnosis and precision medicine. The capacity to capture translation regulation at single-cell [85] and subcellular levels yields deep insights into the spatial dynamics of mRNA translation within the TME. Such insights enable the identification of distinct translational profiles at various metastasis stages. Focusing on how breast cancer cells strategically localize RNA molecules and adapt their translational machinery in response to environmental stresses and oncogenic signals opens avenues for developing targeted therapies and diagnostic tools. These tools can be designed to address the dynamic and heterogeneous nature of tumors specifically. Advancements in comprehending spatial and temporal translation regulation in breast cancer cells could enhance our ability to identify breast cancer-specific molecular signatures, leading to personalized therapeutic approaches. These approaches, tailored to individual tumor profiles, promise to improve treatment efficacy and reduce resistance to existing breast cancer therapies.

Additionally, the growing insights into translational adaptation in cancer cell plasticity (Box 2) are reshaping our comprehension of a tumor's rapid adaptation to dynamic microenvironments and

Box 2. Translational regulation in cancer cell plasticity

Recent research has highlighted phenotypic plasticity as a pivotal new hallmark of cancer, emerging as a key driver of cancer progression [86]. Cancer cell plasticity, defined as the ability of cells to alter their phenotypes in response to environmental cues without genetic mutations, is a catalyst for significant tumor heterogeneity. It originates from a limited number of initial events but experiences substantial phenotypic variation during the course of cancer progression. This plasticity is chiefly a result of the tumor's adaptation to its dynamic microenvironments, which are shaped by various elements, including limited nutrient availability, and the impact of cancer treatment drugs [99,100]. A critical non-genetic mechanism facilitating this plasticity is translation reprogramming [101]. This process, driven by upstream oncogenic signaling pathways, selectively translates oncogenic mRNAs. This activity can significantly influence the expression and function of crucial translation machinery components, thereby further advancing the progression of breast cancer.

In breast cancer, environmental stressors, including hypoxia and mTOR inhibition, specifically activate the translation of 5' UTR isoforms of mRNAs such as NANOG, SNAI1, and NODAL. The proteins produced from these transcripts promote cellular shifts towards stem cell-like phenotypes, fueling metastasis and reducing drug effectiveness [102]. In this context, the mTORC1 complex is crucial, enabling the release of eIF4E from inhibitory proteins. The phosphorylation of eIF4E within the TME is particularly significant for metastatic progression, as it enhances the translation of mRNAs associated with invasion and metastasis [103]. The insights gained from studying translational reprogramming in the context of breast cancer phenotypic plasticity have the potential to enhance our comprehension of the disease's multifaceted nature.

its inherent heterogeneity. These advancements hold the promise of fostering groundbreaking and more effective treatment strategies. Moreover, by unraveling the distinct translational signatures characteristic of individual tumors, it becomes possible to tailor treatments. These customized approaches can precisely target and neutralize the specific pathways driving breast cancer progression in each patient, offering a more personalized and potentially more successful therapeutic strategy.

Concluding remarks and future perspectives

As evidence mounts, the critical role of translational adaptation in breast cancer metastasis becomes increasingly clear. This review has highlighted the collective impact of various elements on the breast cancer translational landscape and their influence on metastatic processes. However, our current understanding predominantly relies on *in vitro* models, such as breast cancer cell lines, constrained not only by the complexity and high input demands but also by the stringent quality requirements of experimental materials. Consequently, there is a critical need for a more systematic understanding of translational responses during the metastatic cascade (see [Outstanding questions](#)), incorporating *in vivo* models and patient samples. Future research should harness these insights to develop targeted therapies that disrupt the adaptive processes in breast cancer cells. With a focus on personalized medicine, exploring individual differences in translational adaptation offers a promising pathway to more effective breast cancer treatments, especially in combating drug resistance and enhancing precision medicine strategies for customized breast cancer therapies.

Acknowledgments

We apologize to those whose work could not be included due to space limitations. H.G. is an Era of Hope Scholar (W81XWH-2210121) and supported by grants from NCI (R01CA240984 and R01CA244634). A.N. is supported by the ATIP-Avenir program, ARC Foundation for Cancer Research, PSL Young Researcher Starting Grant, and Pink Ribbon Award.

Declaration of interests

No interests are declared.

References

- Wilkinson, L. and Gathani, T. (2022) Understanding breast cancer as a global health concern. *Br. J. Radiol.* 95, 20211033
- Ganesh, K. and Massague, J. (2021) Targeting metastatic cancer. *Nat. Med.* 27, 34–44
- Ingolia, N.T. *et al.* (2009) Genome-wide analysis in vivo of translation with nucleotide resolution using ribosome profiling. *Science* 324, 218–223
- Gong, C. *et al.* (2020) Phosphorylation independent eIF4E translational reprogramming of selective mRNAs determines tamoxifen resistance in breast cancer. *Oncogene* 39, 3206–3217
- Kovalski, J.R. *et al.* (2022) Protein synthesis control in cancer: selectivity and therapeutic targeting. *EMBO J.* 41, e109823
- Hiller, D.J. *et al.* (2009) Predictive value of eIF4E reduction after neoadjuvant therapy in breast cancer. *J. Surg. Res.* 156, 265–269
- Tuval-Kochen, L. *et al.* (2013) Eukaryotic initiation factor 2alpha—a downstream effector of mammalian target of rapamycin—modulates DNA repair and cancer response to treatment. *PLoS One* 8, e77260
- Gay, D.M. *et al.* (2022) Translational control through ribosome heterogeneity and functional specialization. *Trends Biochem. Sci.* 47, 66–81
- Moshitch-Moshkovitz, S. *et al.* (2022) The epitranscriptome toolbox. *Cell* 185, 764–776
- Marcel, V. *et al.* (2020) Ribosomal RNA 2'O-methylation as a novel layer of inter-tumour heterogeneity in breast cancer. *NAR Cancer* 2, zcaa036
- Jansson, M.D. *et al.* (2021) Regulation of translation by site-specific ribosomal RNA methylation. *Nat. Struct. Mol. Biol.* 28, 889–899
- Ebright, R.Y. *et al.* (2020) Deregulation of ribosomal protein expression and translation promotes breast cancer metastasis. *Science* 367, 1468–1473
- Wang, W. *et al.* (2014) IRP2 regulates breast tumor growth. *Cancer Res.* 74, 497–507
- Ali, A. *et al.* (2021) Ferritin heavy chain (FTH1) exerts significant antigrowth effects in breast cancer cells by inhibiting the expression of c-MYC. *FEBS Open Bio.* 11, 3101–3114
- Carvalho, J. *et al.* (2020) G-quadruplex, friend or foe: the role of the G-quartet in anticancer strategies. *Trends Mol. Med.* 26, 848–861
- Yu, A.M. *et al.* (2020) RNA drugs and RNA targets for small molecules: principles, progress, and challenges. *Pharmacol. Rev.* 72, 862–898
- Akirtava, C. *et al.* (2022) False-positive IRESes from Hoxa9 and other genes resulting from errors in mammalian 5' UTR annotations. *Proc. Natl. Acad. Sci. U. S. A.* 119, e2122170119
- Jayaram, D.R. *et al.* (2021) Unraveling the hidden role of a uORF-encoded peptide as a kinase inhibitor of PKCs. *Proc. Natl. Acad. Sci. U. S. A.* 118, e2018899118
- Chen, H.H. and Tam, W.Y. (2019) uORF-mediated translational control: recently elucidated mechanisms and implications in cancer. *RNA Biol.* 16, 1327–1338
- Shu, X.E. *et al.* (2020) Nutrient control of mRNA translation. *Annu. Rev. Nutr.* 40, 51–75
- Nomura, Y. and Dohmae, N. (2021) Discovery of a small protein-encoding cis-regulatory overlapping gene of the tumor suppressor gene Scribble in humans. *Commun. Biol.* 4, 1098
- Sharma, Y. *et al.* (2019) A pan-cancer analysis of synonymous mutations. *Nat. Commun.* 10, 2569

Outstanding questions

How do translational responses differ across normal cells, primary breast cancer cells, and metastatic breast cancer cells and are these responses essential for the differentiation between these stages?

How does translational adaptation interact with other stages of gene expression, such as transcriptional regulation, to contribute to breast cancer metastasis?

How does cellular plasticity arise and evolve through translation control as a non-genetic mechanism?

Given the significance of translational adaptation in breast cancer, can we develop tailored RNA medicines to counter drug resistance and refine precision medicine strategies in breast cancer treatment?

23. Supek, F. *et al.* (2014) Synonymous mutations frequently act as driver mutations in human cancers. *Cell* 156, 1324–1335
24. Qian, W. *et al.* (2012) Balanced codon usage optimizes eukaryotic translational efficiency. *PLoS Genet.* 8, e1002603
25. Bae, H. and Coller, J. (2022) Codon optimality-mediated mRNA degradation: Linking translational elongation to mRNA stability. *Mol. Cell* 82, 1467–1476
26. Morales-Polanco, F. *et al.* (2022) Cotranslational mechanisms of protein biogenesis and complex assembly in eukaryotes. *Annu. Rev. Biomed. Data Sci.* 5, 67–94
27. Chen, S. *et al.* (2017) Codon-resolution analysis reveals a direct and context-dependent impact of individual synonymous mutations on mRNA level. *Mol. Biol. Evol.* 34, 2944–2958
28. Shen, X. *et al.* (2022) Synonymous mutations in representative yeast genes are mostly strongly non-neutral. *Nature* 606, 725–731
29. Lampson, B.L. *et al.* (2013) Rare codons regulate KRas oncogenesis. *Curr. Biol.* 23, 70–75
30. Fu, J. *et al.* (2018) Codon usage regulates human KRAS expression at both transcriptional and translational levels. *J. Biol. Chem.* 293, 17929–17940
31. Naso, F.D. *et al.* (2021) Nuclear localisation of Aurora-A: its regulation and significance for Aurora-A functions in cancer. *Oncogene* 40, 3917–3928
32. Cacioppo, R. *et al.* (2023) Differential translation of mRNA isoforms underlies oncogenic activation of cell cycle kinase Aurora A. *eLife* 12, RP87253
33. Navickas, A. *et al.* (2023) An mRNA processing pathway suppresses metastasis by governing translation control from the nucleus. *Nat. Cell Biol.* 25, 892–903
34. He, P.C. and He, C. (2021) m(6) A RNA methylation: from mechanisms to therapeutic potential. *EMBO J.* 40, e105977
35. Huang, H. *et al.* (2020) m(6)A modification in coding and non-coding RNAs: roles and therapeutic implications in cancer. *Cancer Cell* 37, 270–288
36. Zeng, C. *et al.* (2020) Roles of METTL3 in cancer: mechanisms and therapeutic targeting. *J. Hematol. Oncol.* 13, 117
37. Li, Y. *et al.* (2022) METTL3 acetylation impedes cancer metastasis via fine-tuning its nuclear and cytosolic functions. *Nat. Commun.* 13, 6350
38. Barbieri, I. and Kouzarides, T. (2020) Role of RNA modifications in cancer. *Nat. Rev. Cancer* 20, 303–322
39. Yan, H. and Bu, P. (2021) Non-coding RNA in cancer. *Essays Biochem.* 65, 625–639
40. Pinzaru, A.M. and Tavazoie, S.F. (2023) Transfer RNAs as dynamic and critical regulators of cancer progression. *Nat. Rev. Cancer* 23, 746–761
41. Liu, Y. *et al.* (2021) Synonymous but not silent: the codon usage code for gene expression and protein folding. *Annu. Rev. Biochem.* 90, 375–401
42. Goodarzi, H. *et al.* (2016) Modulated expression of specific tRNAs drives gene expression and cancer progression. *Cell* 165, 1416–1427
43. Earnest-Noble, L.B. *et al.* (2022) Two isoleucyl tRNAs that decode synonymous codons divergently regulate breast cancer metastatic growth by controlling translation of proliferation-regulating genes. *Nat. Cancer* 3, 1484–1497
44. Gingold, H. *et al.* (2014) A dual program for translation regulation in cellular proliferation and differentiation. *Cell* 158, 1281–1292
45. Pataskar, A. *et al.* (2022) Tryptophan depletion results in tryptophan-to-phenylalanine substituents. *Nature* 603, 721–727
46. Han, J.M. *et al.* (2012) Leucyl-tRNA synthetase is an intracellular leucine sensor for the mTORC1-signaling pathway. *Cell* 149, 410–424
47. Passarelli, M.C. *et al.* (2022) Leucyl-tRNA synthetase is a tumour suppressor in breast cancer and regulates codon-dependent translation dynamics. *Nat. Cell Biol.* 24, 307–315
48. Goodarzi, H. *et al.* (2015) Endogenous tRNA-derived fragments suppress breast cancer progression via YBX1 displacement. *Cell* 161, 790–802
49. Oliveto, S. *et al.* (2017) Role of microRNAs in translation regulation and cancer. *World J. Biol. Chem.* 8, 45–56
50. Huh, D. *et al.* (2021) A stress-induced tyrosine-tRNA depletion response mediates codon-based translational repression and growth suppression. *EMBO J.* 40, e106696
51. Stone, J.K. *et al.* (2019) Hypoxia induces cancer cell-specific chromatin interactions and increases MALAT1 expression in breast cancer cells. *J. Biol. Chem.* 294, 11213–11224
52. Miao, H. *et al.* (2019) A long noncoding RNA distributed in both nucleus and cytoplasm operates in the PYCARD-regulated apoptosis by coordinating the epigenetic and translational regulation. *PLoS Genet.* 15, e1008144
53. Liu, Y. *et al.* (2021) Long non-coding RNA NR2F1-AS1 induces breast cancer lung metastatic dormancy by regulating NR2F1 and DeltaNp63. *Nat. Commun.* 12, 5232
54. Zheng, C. *et al.* (2023) CRISPR/Cas9 screen uncovers functional translation of cryptic lncRNA-encoded open reading frames in human cancer. *J. Clin. Invest.* 133, e159940
55. Chen, L.L. (2020) The expanding regulatory mechanisms and cellular functions of circular RNAs. *Nat. Rev. Mol. Cell Biol.* 21, 475–490
56. Liu, C.X. and Chen, L.L. (2022) Circular RNAs: characterization, cellular roles, and applications. *Cell* 185, 2016–2034
57. Chen, C.K. *et al.* (2021) Structured elements drive extensive circular RNA translation. *Mol. Cell* 81, 4300–4318
58. Chen, R. *et al.* (2023) Engineering circular RNA for enhanced protein production. *Nat. Biotechnol.* 41, 262–272
59. Qu, L. *et al.* (2022) Circular RNA vaccines against SARS-CoV-2 and emerging variants. *Cell* 185, 1728–1744
60. Ramat, A. and Simoneg, M. (2021) Functions of PIWI proteins in gene regulation: new arrows added to the piRNA quiver. *Trends Genet.* 37, 188–200
61. Ramat, A. *et al.* (2020) The PIWI protein Aubergine recruits eIF3 to activate translation in the germ plasm. *Cell Res.* 30, 421–435
62. Garcia-Borja, E. *et al.* (2024) Critical appraisal of the piRNA-PIWI axis in cancer and cancer stem cells. *Biomark. Res.* 12, 15
63. Wang, C. and Lin, H. (2021) Roles of piRNAs in transposon and pseudogene regulation of germline mRNAs and lncRNAs. *Genome Biol.* 22, 27
64. Tan, L. *et al.* (2019) PIWI-interacting RNA-36712 restrains breast cancer progression and chemoresistance by interaction with SEPW1 pseudogene SEPW1P RNA. *Mol. Cancer* 18, 9
65. El-Naggar, A.M. and Sorensen, P.H. (2018) Translational control of aberrant stress responses as a hallmark of cancer. *J. Pathol.* 244, 650–666
66. Zhang, H. *et al.* (2015) Involvement of Ras GTPase-activating protein SH3 domain-binding protein 1 in the epithelial-to-mesenchymal transition-induced metastasis of breast cancer cells via the Smad signaling pathway. *Oncotarget* 6, 17039–17053
67. Alam, U. and Kennedy, D. (2019) G3BP1 and G3BP2 regulate translation of interferon-stimulated genes: IFITM1, IFITM2 and IFITM3 in the cancer cell line MCF7. *Mol. Cell. Biochem.* 459, 189–204
68. Dermit, M. *et al.* (2020) Subcellular mRNA localization regulates ribosome biogenesis in migrating cells. *Dev. Cell* 55, 298–313
69. Zeng, H. *et al.* (2023) Spatially resolved single-cell translomics at molecular resolution. *Science* 380, eadd3067
70. Reinhardt, K. *et al.* (2022) PIK3CA-mutations in breast cancer. *Breast Cancer Res. Treat.* 196, 483–493
71. Saxton, R.A. and Sabatini, D.M. (2017) mTOR signaling in growth, metabolism, and disease. *Cell* 168, 960–976
72. Xu, Y. and Ruggero, D. (2020) The role of translation control in tumorigenesis and its therapeutic implications. *Ann. Rev. Cancer Biol.* 4, 437–457
73. Frankowski, K.J. *et al.* (2018) Metarrestin, a perinuclear compartment inhibitor, effectively suppresses metastasis. *Sci. Transl. Med.* 10, eaap8307
74. Jahani, Z. and Davoodi, J. (2022) Opposing effects of metformin mediated mTORC1 inhibition on IRES possessing anti-apoptotic proteins in breast cancer cell lines. *Biochem. Biophys. Res. Commun.* 629, 71–77
75. Cao, Y. *et al.* (2023) RNA-based translation activators for targeted gene upregulation. *Nat. Commun.* 14, 6827
76. Metkar, M. *et al.* (2024) Tailor made: the art of therapeutic mRNA design. *Nat. Rev. Drug Discov.* 23, 67–83
77. Mulrone, T.E. *et al.* (2024) N(1)-methylpseudouridylation of mRNA causes +1 ribosomal frameshifting. *Nature* 625, 189–194

78. Kawamura, E. *et al.* (2020) Validation of gene therapy for mutant mitochondria by delivering mitochondrial RNA using a MITO-porter. *Mol. Ther. Nucleic Acids* 20, 687–698
79. Albers, S. *et al.* (2023) Engineered tRNAs suppress nonsense mutations in cells and in vivo. *Nature* 618, 842–848
80. Anastassiadis, T. and Kohrer, C. (2023) Ushering in the era of tRNA medicines. *J. Biol. Chem.* 299, 105246
81. Loayza-Puch, F. *et al.* (2016) Tumour-specific proline vulnerability uncovered by differential ribosome codon reading. *Nature* 530, 490–494
82. Park, M.C. *et al.* (2012) Secreted human glycyl-tRNA synthetase implicated in defense against ERK-activated tumorigenesis. *Proc. Natl. Acad. Sci. U. S. A.* 109, E640–E647
83. Yoon, S. *et al.* (2018) Discovery of novel leucyladenylate sulfamate surrogates as leucyl-tRNA synthetase (LRS)-targeted mammalian target of rapamycin complex 1 (mTORC1) inhibitors. *Bioorg. Med. Chem.* 26, 4073–4079
84. Barczak, W. *et al.* (2023) Long non-coding RNA-derived peptides are immunogenic and drive a potent anti-tumour response. *Nat. Commun.* 14, 1078
85. Vaninsberghe, M. *et al.* (2021) Single-cell Ribo-seq reveals cell cycle-dependent translational pausing. *Nature* 597, 561–565
86. Hanahan, D. (2022) Hallmarks of cancer: new dimensions. *Cancer Discov.* 12, 31–46
87. Suresh, S. and O'Donnell, K.A. (2021) Translational control of immune evasion in cancer. *Trends Cancer* 7, 580–582
88. Cai, X. *et al.* (2021) Current progress and future perspectives of immune checkpoint in cancer and infectious diseases. *Front. Genet.* 12, 785153
89. Han, Y. *et al.* (2020) PD-1/PD-L1 pathway: current researches in cancer. *Am. J. Cancer Res.* 10, 727–742
90. Hu, X. *et al.* (2021) The JAK/STAT signaling pathway: from bench to clinic. *Signal Transduct. Target. Ther.* 6, 402
91. Xue, C. *et al.* (2023) Evolving cognition of the JAK-STAT signaling pathway: autoimmune disorders and cancer. *Signal Transduct. Target. Ther.* 8, 204
92. Chen, B. *et al.* (2022) DENR controls JAK2 translation to induce PD-L1 expression for tumor immune evasion. *Nat. Commun.* 13, 2059
93. Xu, Y. *et al.* (2019) Translation control of the immune checkpoint in cancer and its therapeutic targeting. *Nat. Med.* 25, 301–311
94. Huang, D. *et al.* (2023) Tumour circular RNAs elicit anti-tumour immunity by encoding cryptic peptides. *Nature* 625, 593–602
95. Amaya, L. *et al.* (2023) Circular RNA vaccine induces potent T cell responses. *Proc. Natl. Acad. Sci. U. S. A.* 120, e2302191120
96. Araki, K. *et al.* (2017) Translation is actively regulated during the differentiation of CD8(+) effector T cells. *Nat. Immunol.* 18, 1046–1057
97. Wolf, T. *et al.* (2020) Dynamics in protein translation sustaining T cell preparedness. *Nat. Immunol.* 21, 927–937
98. Tan, T.C.J. *et al.* (2022) Translation factor eIF5a is essential for IFN γ production and cell cycle regulation in primary CD8(+) T lymphocytes. *Nat. Commun.* 13, 7796
99. Lee, L.J. *et al.* (2021) Cancer plasticity: the role of mRNA translation. *Trends Cancer* 7, 134–145
100. Gupta, P.B. *et al.* (2019) Phenotypic plasticity: driver of cancer initiation, progression, and therapy resistance. *Cell Stem Cell* 24, 65–78
101. Fabbri, L. *et al.* (2021) The plasticity of mRNA translation during cancer progression and therapy resistance. *Nat. Rev. Cancer* 21, 558–577
102. Jewer, M. *et al.* (2020) Translational control of breast cancer plasticity. *Nat. Commun.* 11, 2498
103. Robichaud, N. *et al.* (2018) Translational control in the tumor micro-environment promotes lung metastasis: phosphorylation of eIF4E in neutrophils. *Proc. Natl. Acad. Sci. U. S. A.* 115, E2202–E2209